

Module 38 - Array Methods (concat, slice, flat & fill)

1. concat() Method

The concat() method is used to combine two or more arrays and returns a new array without changing the original arrays.

Example Code:

```
let arr = [1, 2, 3, 4];
let arr1 = [4, 5, 6, 7];
// Combines two arrays and returns a new array.
let newArr = arr.concat(arr1);
// Adds values to the end and returns a new array.
let newArr1 = arr.concat(10, 20, 30);
// Creates a shallow copy of the array.
let newArr2 = [].concat(arr);
arr[0] = 111;
console.log(newArr, newArr1, newArr2, arr);
```

2. slice() Method

The slice() method is used to extract a portion of an array and returns a new array without changing the original array.

Example Code:

```
let newArray = [10, 2, 3, 4, 5, 6, 7];
// Creates a copy of the entire array.
let slicedVal = newArray.slice();
newArray[0] = 101;
// Extracts elements from index 1 to the end.
let slicedVal1 = newArray.slice(1);
// Extracts elements from index 1 to index 3.
let slicedVal2 = newArray.slice(1, 4); // (start, end - 1) (1, 4 - 1 = 3)
// Extracts elements from index 0 to index 2.
let slicedVal3 = newArray.slice(0, 3); // (start, end - 1) (0, 3 - 1 = 2)
console.log(slicedVal, slicedVal1, slicedVal2, slicedVal3);
```

3. flat() Method

flat() Method is used to convert nested arrays into a single-level array by removing the specified level of nesting.

Example Code:

```
let Arr = [1, 2, 3, [4, [5, 6, [40, 50, [70, 80]]]]];  
// Flattens the array up to depth 2.  
let NewArr = Arr.flat(2);  
// Flattens all nested arrays into a single array.  
let NewArr1 = Arr.flat(Infinity);  
console.log(Arr, NewArr, NewArr1);
```

4. fill() Method

The fill() method is used to replace elements of an array with a specified value from a given start index to an end index.

Example Code:

```
let ARR = [10, 20, 30];  
// Fills the array with 101 from index 0 to index 1.  
ARR.fill(101, 0, 2); // (value, start, end - 1) (101, 0, 2 - 1 = 1)  
// Fills the array with 103 at index 2.  
ARR.fill(103, 2, 3); // (value, start, end - 1) (103, 2, 3 - 1 = 2)  
console.log(ARR);
```